RL-Based Active Suspension Control Using a Quarter-Car Simulation

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GitHub: `https://github.com/CrazAzartto/active-suspension-rl`

***Abstract*—Passive car suspensions trade ride comfort against road holding. We cast active quarter-car suspension as continuous-control RL: six sensor observations, one bounded force action, a quadratic reward, trained with PPO. On held-out roads PPO cuts body acceleration 22–43% versus passive but worsens tyre-load variation.**

***Index Terms*—reinforcement learning, PPO, active suspension, quarter-car model, ride comfort, road holding.**

I. Problem and Motivation

A passive suspension (spring k_s, damper c_s) trades *ride comfort* (low body acceleration) against *road holding* (small dynamic tyre-load variation) and bounded *suspension travel*. An active suspension adds a force actuator, but must act on an unknown random road with limited sensing and a saturating actuator. We pose this as a partially observed MDP and learn a policy with Proximal Policy Optimization (PPO) [1]. The simulator, environment and evaluation code were written for this project; only NumPy, Gymnasium [3] and Stable-Baselines3 [2] are used as libraries, and no existing RL repository was copied.

II. Quarter-Car Dynamics

With z up and displacements from static equilibrium (gravity cancels), body z_s, wheel z_u, road z_r, actuator force u:

$$m_s\ddot{z}_s = -k_s(z_s - z_u) - c_s(\dot{z}_s - \dot{z}_u) + u, \tag{1}$$

$$m_u\ddot{z}_u = k_s(z_s - z_u) + c_s(\dot{z}_s - \dot{z}_u) - k_t(z_u - z_r) - u. \tag{2}$$

Parameters (plausible round values, *not* identified from a real car): m_s=290 kg, m_u=59 kg, k_s=16 kN/m, c_s=1 kN s/m, k_t=190 kN/m (coupled modes 1.14 Hz and 9.34 Hz, damping ratios 0.21 and 0.15). Integration: fixed-step RK4, 1 ms, force held constant over each 10 ms control step; episode 10 s (1000 steps). Roads are C^1 profiles at constant speed, $z_r(t)=z(vt)$: flat, raised-cosine bump, and ISO 8608-style random roads [4] (classes A–E, $G_d(n)=G_d(n_0)(n/n_0)^{-2}$, 100 random-phase cosines). Assumptions: linear spring/damper/tyre, no bump stops, no tyre lift-off (tyre-load ratio $1+F_t/((m_s+m_u)g) \le 0$ is only *flagged*), ideal actuator without lag.

III. RL Formulation

A. State (observation) space $\mathcal{S}$

$o_k \in \mathbb{R}^6$ (dimension 6; fixed normalisation, no clipping):

$$o_k = \left[\frac{\ddot{z}_s}{2},\ \frac{z_s - z_u}{0.05},\ \frac{\dot{z}_s - \dot{z}_u}{0.5},\ \frac{z_u - z_r}{0.01},\ \frac{\dot{z}_u}{0.5},\ a_{k-1}\right].$$

Road velocity and the upcoming road are *not* observed, so the problem is partially observed.

B. Action space $\mathcal{A}$

Continuous $a_k \in [-1,1]$, force $u_k = \mathrm{clip}(a_k)\,u_{\max}$, $u_{\max}$=1500 N ($u>0$ pushes the body up and the wheel down).

C. Reward R

With overbars the mean over the 10 sub-steps of a control step, $d=z_s-z_u$, $F_t=k_t(z_u-z_r)$:

$$r_k = -\left[w_a\frac{\overline{\ddot{z}_s^2}}{a_{\mathrm{ref}}^2} + w_d\frac{\overline{d^2}}{d_{\mathrm{ref}}^2} + w_t\frac{\overline{F_t^2}}{F_{\mathrm{ref}}^2} + w_u a_k^2 + w_\Delta\left(\frac{a_k - a_{k-1}}{2}\right)^2\right], \tag{3}$$

$(w_a, w_d, w_t, w_u, w_\Delta)=(1, 0.3, 0.3, 0.05, 0.05)$, a_{ref}=1 m/s^2, d_{ref}=0.015 m, F_{ref}=$0.25(m_s+m_u)g$=856 N. The scales are the rounded RMS response of the passive car on ISO C at 15 m/s, so each term is $O(1)$ and the weights express priority (comfort first). Safety-guard termination ($|d|>0.30$ m, $|z_u-z_r|>0.15$ m, non-finite state) costs 10 per remaining step; it never occurred (0 of 113 PPO evaluation episodes). Weights are a first choice, *not* tuned.

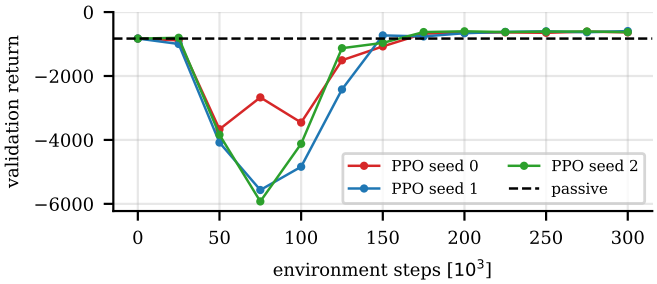


Fig. 1. Validation return (mean over 6 validation roads) for three PPO training seeds vs passive on the same roads (higher is better). Best: -612, -594, -602; passive: -829.

IV. Environment and PPO Training

The Gymnasium environment resamples road, speed and seed at each reset. **Training mixture:** flat (10–20 m/s); bump ($h\in[0.02, 0.04]$ m, $L\in[1.5, 3]$ m, 8–14 m/s); ISO B and C (10–20 m/s). PPO suits the bounded continuous force, the cheap simulator and needs stable updates (clipped objective). Stable-Baselines3 setup: MLP actor/critic 64×64 tanh; 4 envs × 1000-step rollouts; batch 500; 10 epochs; learning rate 3×10^{-4}; γ=0.99; GAE λ=0.95; clip 0.2; entropy coef. 0; seeds 0, 1, 2 (seed 0 in Table I); 300 000 steps per seed (139 s, one CPU core). Only the reward is normalised (running statistics). Every 25k steps the policy is validated on six fixed seeds (90000–90005) of the training mixture; the best checkpoint (275k steps) is kept. Evaluation seeds ($\ge$100000) are never used for training or selection.

V. Baseline and Evaluation Protocol

The **passive baseline** applies $u\equiv0$ on the identical vehicle; c_s=1000 N s/m gave the highest passive reward on ISO B/C in a sweep over 500–4000 (stiffer damping lowers tyre-load variation but raises acceleration). PPO (deterministic) and passive are run on the *same* road, speed, reset seed and initial state (road height asserted identical). In Table I: **T**=training distribution; **U**=unseen road class A or unseen initial offset (body +20 mm); **H**=held-out: taller/longer/faster bump (h=0.055 m, L=3.5 m, 16 m/s), ISO D at 10 m/s, and the 'test' mixture (tall bump; ISO A at 22–28 m/s; ISO D at 8–12 m/s). Random roads use 20–30 seeds, deterministic scenarios 1. Metrics are physical (unweighted RMS acceleration, no ISO 2631 filter; deflection; RMS dynamic tyre load; contact-loss flag; actuator force); return is kept separate. A hand-designed skyhook law $u=-1500\,\dot{z}_s$ [5] (*not RL*, untuned) serves only as a reference.

VI. Results

Fig. 1: in all three seeds the validation return first falls far below passive (down to -5925 at 50–100k steps, while the policy still applies large exploratory forces) and exceeds the passive level by 150–175k steps. **Comfort:** PPO lowers RMS and peak body acceleration in every scenario: 22–26% on random roads (all paired seeds better), 22–43% on the three held-out cases, 49% on the in-distribution bump; the untuned skyhook reference gives about 8.5%. **Trade-offs:** on random roads RMS tyre load is 22–25% *worse* than passive, deflection is worse in most scenarios, and the contact-loss flag on ISO D rises from 1.4% to 4.3% of steps;

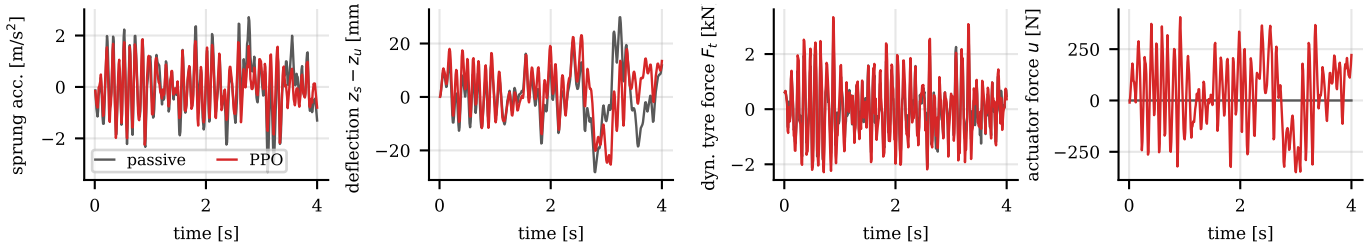


Fig. 2. Passive vs PPO on an ISO-C road, 15 m/s (seed 100000, first 4 s): body acceleration, suspension deflection, dynamic tyre force, and PPO actuator force (passive: 0 N).

TABLE I
PPO VS PASSIVE UNDER MATCHED CONDITIONS (MEANS OVER n EPISODES, SEEDS $\geq$100000)

		RMS acc [m/s^2]		PPO change vs passive [%] (+ = better)				Skyhook	PPO	Contact-loss
Scenario (set)	n	passive	PPO	RMS acc	peak acc	RMS defl	RMS tyre	RMS acc [%]	RMS u [N]	steps [%] pass./PPO
bump 10 m/s (T)	1	0.386	0.197	+49.1	+37.0	+3.3	+2.6	+15.8	68	0.0/0.0
ISO B 15 m/s (T)	20	0.575	0.437	+24.0	+23.9	-18.2	-22.8	+8.5	104	0.0/0.0
ISO C 15 m/s (T)	20	1.149	0.877	+23.6	+21.8	-12.3	-23.6	+8.5	194	0.0/0.1
ISO A 15 m/s (U)	20	0.287	0.218	+24.0	+24.1	-36.3	-22.2	+8.5	61	0.0/0.0
flat, +20 mm start (U)	1	0.131	0.035	+73.6	+70.2	-46.7	+72.3	+37.6	52	0.0/0.0
tall bump 16 m/s (H)	1	0.541	0.309	+43.0	+33.7	+1.2	-5.9	+10.3	88	0.0/0.0
ISO D 10 m/s (H)	20	1.861	1.447	+22.2	+21.0	-8.6	-24.6	+8.3	296	1.4/4.3
test mixture (H)	30	0.925	0.683	+26.2	+25.7	-8.9	-22.1	+9.4	152	0.5/1.4

TABLE II
PPO CHANGE VS PASSIVE [%] (+ = BETTER), MEAN±STD OVER 3 TRAINING SEEDS

Scenario (set)	RMS acc	RMS defl	RMS tyre load
bump 10 m/s (T)	+46.9 ± 1.9	+4.9 ± 1.5	+3.1 ± 1.0
ISO B 15 m/s (T)	+24.1 ± 0.9	−10.7 ± 6.5	−25.0 ± 2.8
ISO C 15 m/s (T)	+23.9 ± 0.2	−5.3 ± 6.0	−24.0 ± 1.6
ISO A 15 m/s (U)	+24.1 ± 1.2	−26.9 ± 8.3	−25.3 ± 3.4
flat, +20 mm (U)	+66.5 ± 6.3	−36.3 ± 10.0	+63.6 ± 7.7
tall bump (H)	+39.9 ± 2.7	+3.3 ± 1.8	−3.5 ± 2.2
ISO D 10 m/s (H)	+22.5 ± 0.6	−2.2 ± 5.6	−23.5 ± 1.4
test mixture (H)	+26.0 ± 0.4	−3.2 ± 4.9	−21.4 ± 1.2

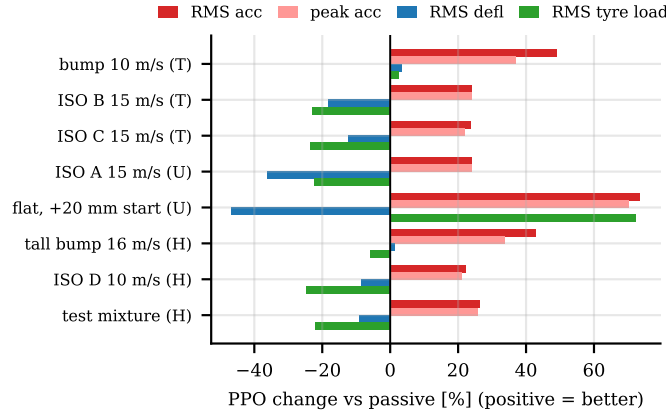


Fig. 3. Change of PPO relative to passive per scenario and metric (from Table I data; positive = better, negative = degradation). Comfort gains come with negative tyre-load and deflection bars on most scenarios.

skyhook leaves tyre load unchanged (+0.2% on ISO C). On the 10 m/s bump tyre load changes by +2.6%. **Effort:** moderate (RMS 52–296 N, peak $\leq$1046 N, saturation 0.0%). **Seed robustness:** retraining with seeds 1 and 2 gives the same picture (Table II): on random roads every seed reduces RMS acceleration by 22–26% and increases RMS tyre load by 20–29%.

VII. Discussion and Conclusion

Mean per-step reward terms on ISO C (passive→PPO) are: acceleration 1.34→0.64, deflection 0.21→0.26, tyre force 0.31→0.48, total 1.86→1.38. PPO thus optimises the stated objective, buying a large comfort gain with higher tyre-force and travel terms (comfort weight 1.0 vs 0.3); a larger w_t should shift this balance (not tested). *Limitations:* only three training seeds (no significance test); a 1M-step run gave no better validation return; untuned weights; unweighted acceleration; linear tyre; ideal actuator; unvalidated parameters; hidden road velocity. A PPO policy trained on a road mixture thus generalised to unseen and held-out roads for comfort, at a clear road-holding cost; we do not claim superiority over passive suspension. **Reproducibility:** code, trained model, result CSVs and the commands to regenerate every table and figure of this report are in the repository (`README.md`); numbers here are generated from the saved CSV files.

References

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